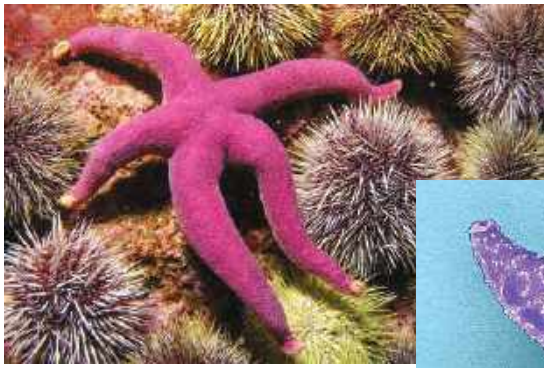


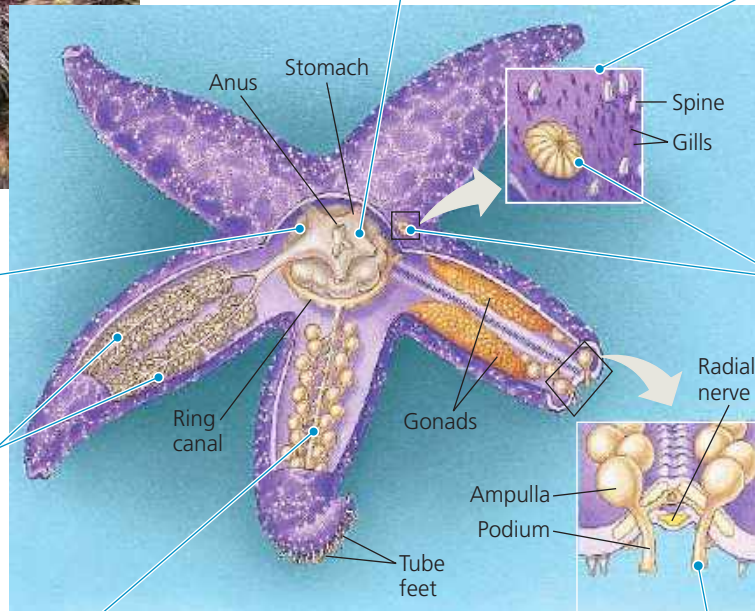


A short digestive tract runs from the mouth on the bottom of the central disk to the anus on top of the disk.

The surface of a sea star is covered by spines that help defend against predators, as well as by small gills that provide gas exchange.

Central disk. The central disk has a nerve ring and nerve cords radiating from the ring into the arms.

Digestive glands secrete digestive juices and aid in the absorption and storage of nutrients.



Madreporite. Water can flow in or out of the water vascular system into the surrounding water through the madreporite.

Radial canal. The water vascular system consists of a ring canal in the central disk and five radial canals, each running in a groove down the entire length of an arm. Branching from each radial canal are hundreds of hollow, muscular tube feet filled with fluid.

Each tube foot consists of a bulb-like ampulla and a podium (foot portion). When the ampulla squeezes, water is forced into the podium, which expands and contacts the substrate. Adhesive chemicals are then secreted from the base of the podium, attaching it to the substrate. To detach the tube foot, de-adhesive chemicals are secreted and muscles in the podium contract, forcing water back into the ampulla and shortening the podium. As it moves, a sea star leaves an observable "footprint" of adhesive material on the substrate.

▲ **Figure 33.43 Anatomy of a sea star, an echinoderm (top view).** In the diagram, the digestive glands, radial canal, and gonads are illustrated in separate arms for clarity, but each is present in every arm. The photograph shows a sea star surrounded by sea urchins, which are also echinoderms.

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members of one genus can even regrow an entire body from a single arm if part of the central disk remains attached.

The clade Asteroidea, to which sea stars belong, also includes a small group of armless species, the *sea daisies*. Only three species of sea daisies are known, all of which live on submerged wood. A sea daisy's body is typically disk-shaped; it has a five-sided organization and measures less than a centimeter in diameter (**Figure 33.44**). The edge of the body is ringed with small spines. Sea daisies absorb nutrients through a membrane that surrounds their body.

► **Figure 33.44 A sea daisy (clade Asteroidea).**



Ophiuroidea: Brittle Stars

Brittle stars have a distinct central disk and long, flexible arms (**Figure 33.45**). They move primarily by lashing their arms in serpentine movements. The base of a brittle star tube foot lacks the flattened disk found in sea stars but does secrete adhesive chemicals. Hence, like sea stars and other

▼ **Figure 33.45 A brittle star (clade Ophiuroidea).**

